geostandards-INDG20-60

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Introduction

Interoperability in the field of geoinformation, from eCH to OGC

Recently, in 2022, the Open Geospatial Consortium (OGC) and the Open Source Geospatial

Foundation (OSGeo) have renewed their Memorandum of Understanding. While the initial agreement dates from 2008, this renewal is especially motivated by the current focus on the OGC APIs and the will to empower developers (even non-geospatial) to leverage location in their development. Indeed, the OGC has initiated the creation of a new generation of standards based on the OpenAPI specifications to facilitate the integration of geospatial data in modern web applications and systems.

Underpinning the OGC Standards Roadmap cite:[ogc-roadmap], the development of all these standards represents a significant number of activities carried out by various OGC working groups, Testbeds and pilots from the OGC Innovation Program cite:[ogc-ip]. Some standards have been approved, many are still under development and it is therefore not always easy to follow their progress. Indeed, while some geodata infrastructures involving national entities as the(e.g. Meteorological Service of Canada cite:[eccc-msc] in Canada) are already deploying this new generation of standards, some initiatives run a phase of experimentation such as for instance the Geonovum OGC API Testbed Platform cite:[apitestbed].

From a practical perspective we can ask the question how organizations and institutions anticipate to leverage this new generation of standards in order to deploy a geospatial data infrastructure. This issue is at the core of this article and at the center of a project that seeks to address the issue by running an OGC API Testbed platform with a special focus to the Swiss context. This project is embedded in the NGDI Resources program related to the Swiss Geoinformation Strategy cite:[indg] with the purpose to contribute to the upcoming revision of Swiss e-government standards regarding geoinformation such as the eCH-0056 cite:[eCH-0056] standard. This project is about a research carried out by academic partners (HEIG-VD, SUPSI, UNIGE) as well as the Swiss Office for Topography Swisstopo.

Swiss National Context

In Switzerland the exchange and publication of geographic in- formation is governed by the Federal Act on Geoinformation (GeoIA) cite:[geoia]. The GeoIA is accompanied by two regulations:

- GeoIV/OGeo (Federal Council Ordinance) cite:[ogeo]
- GeoIV/OGeo-swisstopo (Ordinance specific for the Federal Office of Topography swisstopo) cite:[ogeo-swisstopo]

The GeoIV/OGeo contains the fundamental provisions, while the GeoIV/OGeo-swisstopo regulates the technical details, which can be modified by the competent federal office (swisstopo), with the participation of the cantons and after consultation with different partner organisations:

- eCH, a national association for the establishment of e-government standards
- SOGI/OSIG, a national association for geoinformation
- KGK-CGC, an organization of the Swiss Cantons in the field of geoinformation management

The GeoIV/OGeo-swisstopo specifies criteria such as the geodetic reference system, the description language for modeling geospatial data, metadata standards for geoinformation and minimal requirements for geodata.

In its 7th article: ”Minimal requirements for geoservices”, the GeoIV/OGeo-swisstopo refers to the e-

Government standard eCH-0056 entitle Application Profile for Geoservices cite:[eCH-0056], which defines the implementation of basic geoservices by means of a set of additional guidelines and recommendations which make the services suitable for practical use. In addition, there are other regulations, such as metadata models cite:[gm03] and protocols for the exchange of base geodata under federal law.

As the eCH-0056 document currently refers to the WxS family of OGC specifications (WMS, WMTS, WFS, WCS, CSW, SE, SLD, SOS), our research aims to assess the maturity and the benefits of the new OGC APIs in the perspective of a revision of its legal framework.

Methodology

Given the above general purpose, the intent is to develop knowledge and practice of these OGC API standards so as, for instance, to update guidelines of the instructions stated by eCH- 0056. More generally, how can the advances coming out from the OGC standardization process be integrated into the Swiss standardization process and even vice versa. The idea in this project is to setup a Testbed Platform to connect these two processes as illustrated by figure 1 by considering issues related to how requirements defined by stakeholders are handled by standardization works both sides. By running at each iteration a showcase on specific use cases with a set of standards to challenge, the Testbed Platform brings results as update guidelines (e.g. for eCH working groups in the field of geoinformation) and feedbacks (e.g. OGC Change Requests). In addition, such a vision can provide an opportunity for dissemination and training material creation (e.g. tutorials, workshops).

Several activities are to be carried out to feed and run such a platform, among them: requirements analysis with stakeholders, identification of underlying standards to challenge, subscription/following in the relevant standards working groups, definition of experimental cases and evaluation criteria, deployment and maintenance of software to run showcases, documentation and dissemination.

For this project, to be seen as a first iteration, the expected outcomes include both quantitative and qualitative results that will be compiled into practical e-government guidelines for the implementation of standards from the OGC API family. The selected mainstream topic for the showcase and the underlying experimental cases is about climate change. While not yet connected in a complex pilot study, each case represents some of the required components from sensors to portrayal. The study is organized in three parts to challenge standards with software related to: (1) sensors data (2) data discovery, access and portrayal (3) earth observation data.

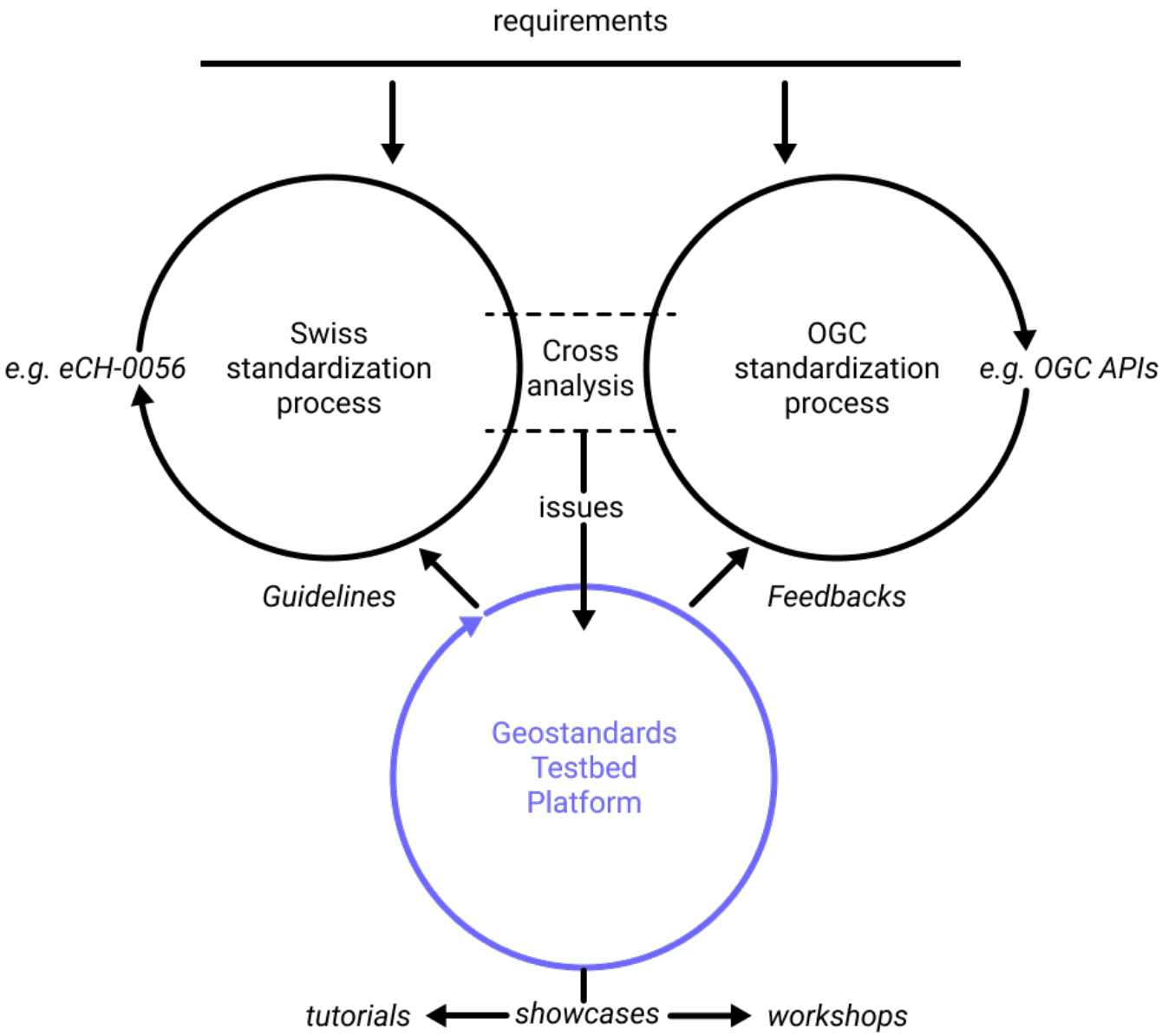


Figure 1. Proposed vision to connect Swiss and OGC standardization processes through a Testbed Platform

In term of software, the idea is to offer a unique project entry point for the discovery, experimentation and analysis of the new OGC APIs. To develop the knowledge and practice required, openness is not only related to the standards evaluated, but also required for the software aspects of the Testbed platform. That's why we identified the Geonovum OGC API Testbed tooling stack cite:[apitestbed] as a good base. It includes already open sourceopen-source packages and provides a detailed documentation available on GitHub. We deployed a project instance at https://ogc.heig-vd.ch/ (figure 2) which still needs to get populated with the underlying data and configured to serve everything required to run the experimental cases, from server to client sides. This solution will also be used to create tutorials in the style of the documentation proposed by Meteo Canada cite:[eccc-msc].

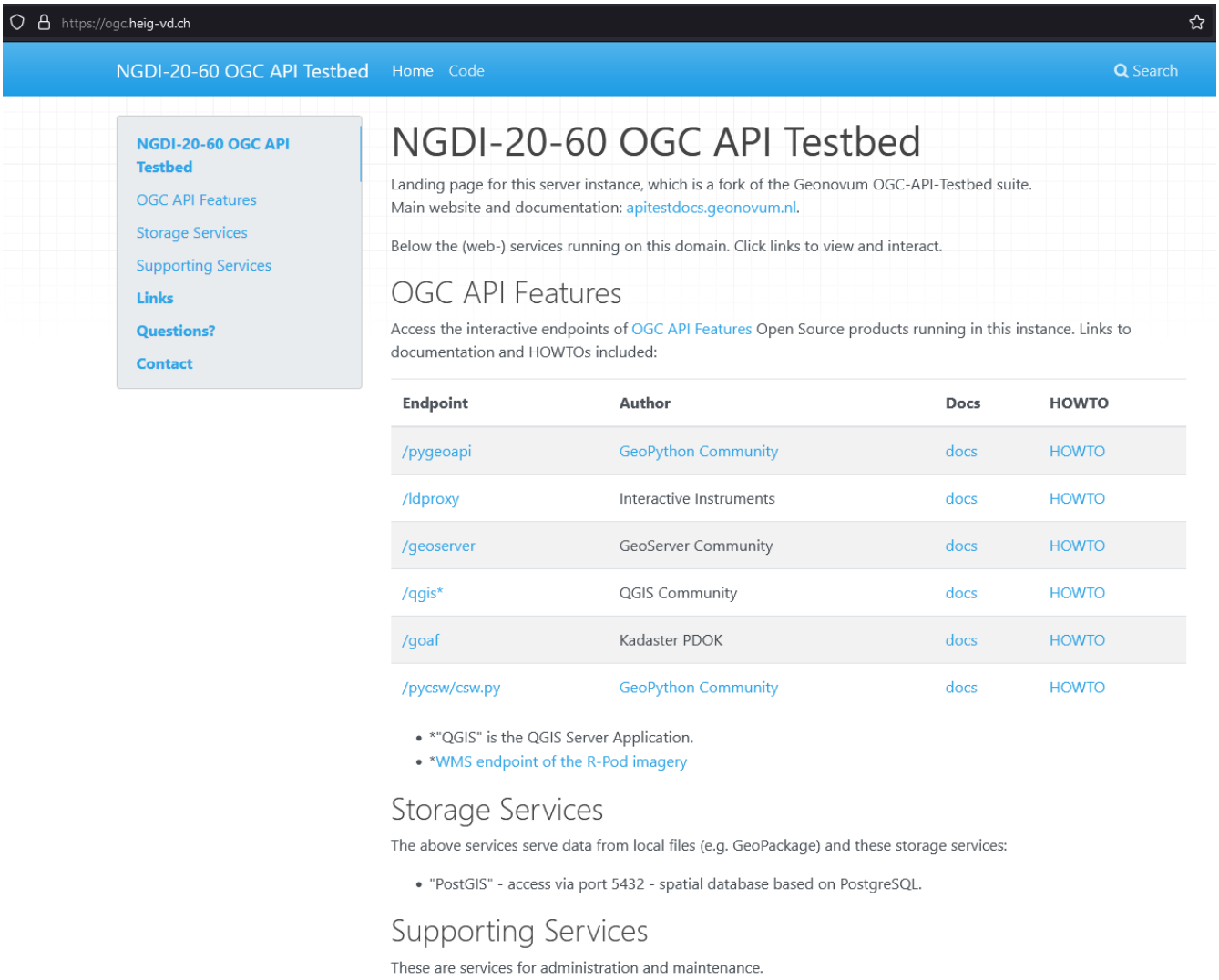


Figure 2. Landing page of the Testbed platform

This solution is expected to bring good visibility toward all actors concerned by the Swiss standardization process. It is to raise interest from government stakeholders, companies and universities, technical people or not, as a community in synergy when considering the advances at the OGC.

To test the experimental cases, FOSS4G implementations are deployed, especially server-side software with:

- FROST : focusing on sensor observation standards
- Geoserver : quite active in the implementation of a wide set of OGC API standards
- ldproxy : focusing on OGC API Features as an adapter sitting in front of existing WFS services
- pygeoapi : active in the implementation with the largest set of OGC API standards of the panel
- QGIS Server : quite active community in Switzerland, implementing also an OGC API Features client

The choice of these solutions was made to cover all the standards to be challenged in this project as well as with the variety of programming languages of the most common OSGeo software, the ease of deployment and the quality of their documentation. Table 1 has been built by going over the OGC API landing page of each, in conformance with the Landing Page Requirements Class of the

specification OGC API - Common - Part 1: Core (OGC, 2021a).

Table 1. Implementations according to conformance declarations (status as of May 2022)

	FROST	Geoserver	ldproxy	pygeoapi	QGIS Server
SensorThingsAPI (OGC:18-062r2)	▯				
Features (OGC:17-069r3)		▯	▯	▯	▯
Maps (OGC:20-058)		▯			
Styles (OGC:20-009)		▯			
Tiles (OGC:20-057)		▯			
Coverages (OGC:19-087, draft)		▯		▯	
DGGS (draft)		▯			
EDR (OGC:19-086r4)				▯	
Processes (OGC:18-062r2)				▯	
Records (OGC:20-004)				▯	

Work Package 2 (WP2)

The Work Package 2 (named hereafter WP2) has the intent to evaluate the current OGC standards related to in-situ observations.

Namely, we'll investigate the:

- Sensor Observation Service (SOS)
- SensorThingsAPI (STA)
- Environmental Data Retrival (EDR)

Motivations

In-situ data collection and distribution provide key information to address several issues, particularly when combined with other data sources like satellite images or socio-economic contextual data. In particular in-situ observations are key complementary data that

support/complete satellite providing data that have high fidelity, resolution and consistency [1].

In 2022, OGC is launching another pilot, the Climate Change Services Initiative, which will run from 2022 through 2026.

[1] *Fekete, B. M., Robarts, R. D., Kumagai, M., Nachtnebel, H. P., Odada, E., & Zhulidov, A. V. (2015). Time for in situ renaissance.*

Approach

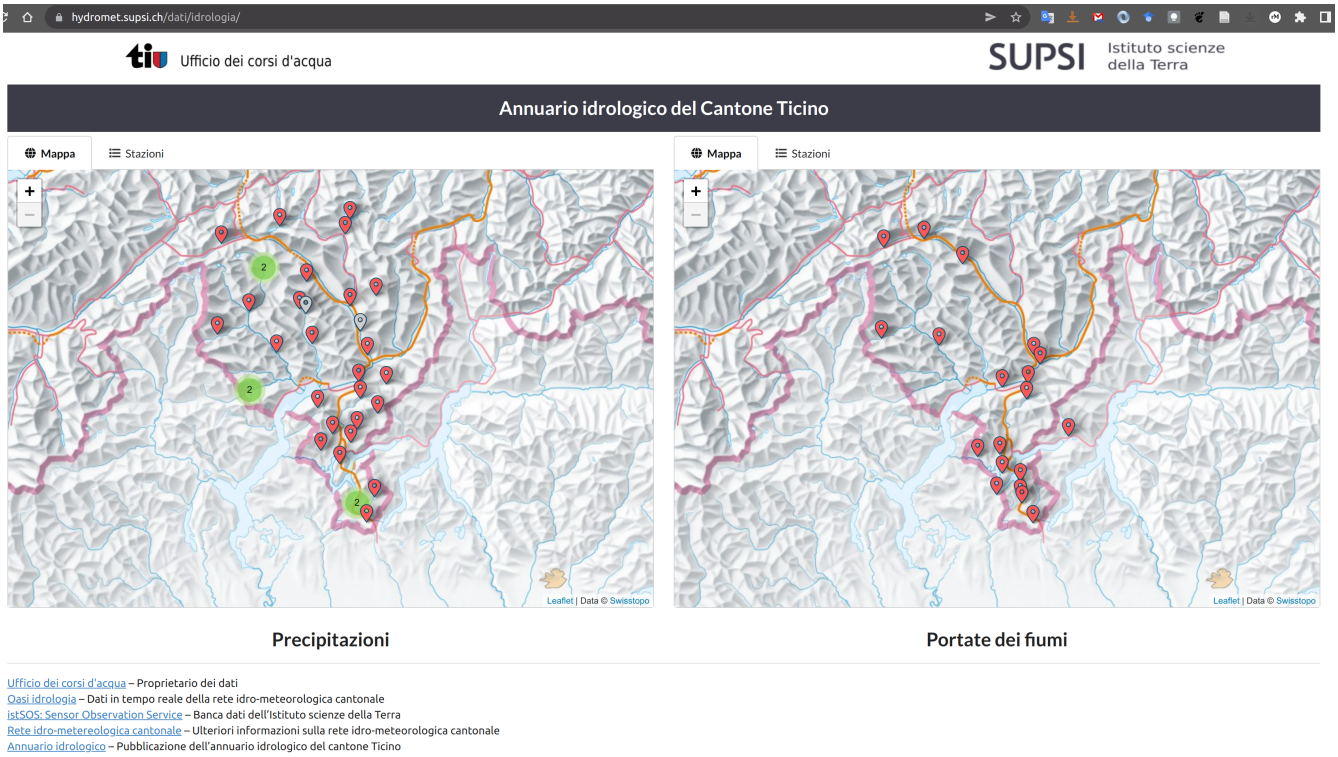
The idea is to set-up experiment and compare the different standards trough different implementations to extrapolate practical considerations and advantages and disadvantages of the different OGC standards.

				2021					2022					
				N	D	J	F	M	A	M	G	J	A	S
		duration	month	1	2	3	4	5	6	7	8	9	10	11
WP2.1	**OGC SensorThings API**													
T2.1.1	Setup ST-API web service													
T.2.1.2	Comparison with OGC SOS													
T2.1.3	Reporting													
WP2.2	**OGC Environmental Data Retrival**													
T2.2.1	Setup an EDR service													
T2.2.2	Comparison with OGC SOS													
T2.2.3	Reporting													
WP2.3	**Observation data analysis tool**													
T2.3.1	Extend OAT to support STA & EDR													
T2.3.2	Use OAT to serve Time Series processing (Case study)													
WP2.4	**Sensor standards solutions banchmark**													
T2.4.1	Benchmark of STA, EDR, SOS													
T2.4.2	Reporting													

Software & standards

- setting up a Sensor Observation Service using istSOS
  - installed at https://geoservice2.ist.supsi.ch/indg/istsos/admin/
- setting up the Sensor Things API using FROST
  - installed at https://geoservice2.ist.supsi.ch/indg/frost/

Dataset



The hydro-meteorological monitoring network of the Canton Ticino (Southern Switzerland), which is currently managed using the SOS standard, has been selected as representative of a practical implementation of basic data required for the climate change impact assessment pipeline. The network, which has a 40 years long time-series, is currently composed of 60 stations and 140 sensors observing precipitation, air temperature and humidity, water temperature, river height. Collected information is operationally used by the local administration to design and actuate water resources protection and allocation to guarantee a sustainable management of the resource and the natural environment while protecting from the impacts of extreme events like floods and droughts. The Sensor Things API operational applicability is evaluated by testing the adopting this standard to fulfill all the major in place daily practical operations like for example data quality management, data sharing with third parties, data collection from vendor specific sensors and data analyses and visualization.

** A second dataset of possible interest consist in the observation of groundwater level and temperature in Canton Ticino. The observations are not sampled at regular time intervals and includes:

- Wells (7'999) with 210'780 observations
- Boreholes (4'664) with 1'440'806 observations

```
    Mapping current DB with istSOS keys and prepare script to migrate data.
**
```

First test migrate automatic sensors

- migrate data to istSOS: DONE
- migrate data to FROST: DOING

Comparison: how SOS fit the

TODOS

- test registering new stations
- test inserting data
- test acquire metadata of sensor
- test acquire data

Questions

1. How modern sensor architecture works?
2. Does the standard fit with the up-to-date approaches?
3. What are the advantages and drawbacks?
    - il sensore è il tipo di sensore (Optod-123) oppure lo specifico strumento con il suo serial number (Otpod-123 S/N: 12345) ?
    - comse si tiene conto delle variazioni dei sensori su una stazione di misura long-term?
    - il datastream è legato a lla serie storica di monitoraggio di un punto di osservazione (a cui può nel tempo cambiare sensori etc..)?
    - il multidatastream è un'osservazione complessa (tipo array) ma non può essere recuperata per ciascuna componente: inserico (ws,wd) e le devo sempre prendere insieme?
    - la location è della Things

Work Package 3 (WP3)

Background

In the context of data discovery, access and portrayal, the wellknown OGC WxS standards WFS cite:[ogc-wfs], WMS cite:[ogc-wms], WMTS cite:[ogc-wmts] have been used for more than ten years and still widely in use. In association with these standards, styling aspects are defined by the standards SLD cite:[ogc-sld] & SE cite:[ogc-se]. These are typically referenced by the eCH-0056 Geoservices Application Profile: WMS 1.3.0 (section 6.7), WMTS 1.0.0 (section 6.8), WFS 2.0 (section 6.9.2), WCS 2.0.1 (section 6.9.3), CSW 2.0.2 (section 6.10), SE 1.1.0 (section 6.11) and SLD 1.1.0 (section 6.12).

For this project part, we focus on standardisation work at the OGC related to discovery, data access to visualisation, as made available at https://ogcapi.ogc.org/ and according to their versioning mentioned by Table 2. Indeed, the table describes the relationship between the considered OGC APIs and their current equivalents in the context of raster and vector related standards.

Table 2. From WxS family to OGC APIs

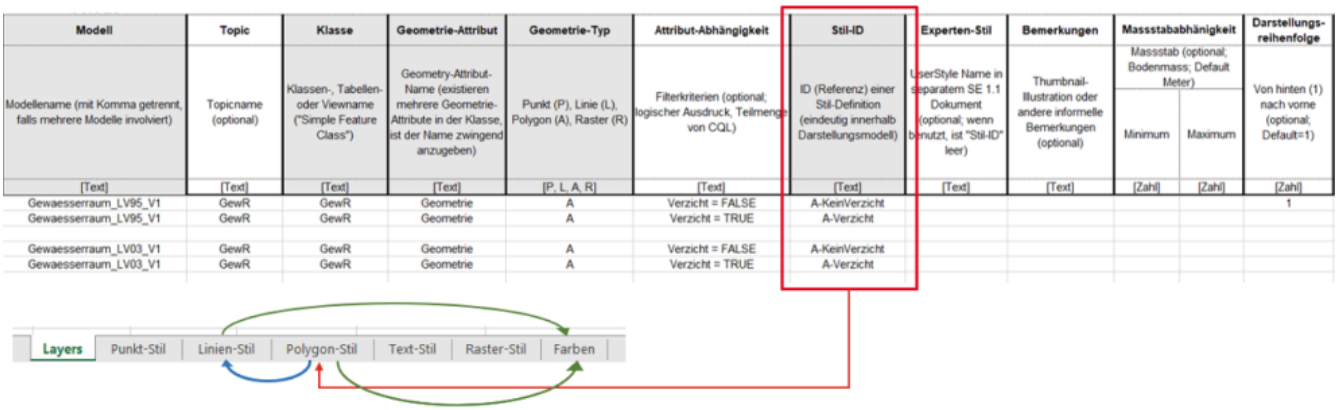
OGC API	Version	WxS fashioned

OGC API Features	1.0	WFS
OGC API Maps	0.0.1	WMS
OGC API Styles	1.0.0	SLD
OGC SymCore	1.0	SE
OGC API Tiles	0.0.4	WMTS
OGC API Records	1.0.0	CSW

Testbed actions

To test and analyze these standards and specifications, two experimental cases are setup:

- with the use of Geoclimate cite:[Bocher2021], an open source geospatial toolbox that computes a set of urban climate parameters based on OpenStreetMap data. The intent is to publish these parameters with metadata, data and maps using the new OGC APIs cite:[ogc-api]
- with the Swiss National geodata models that have been published by the Swiss Government as Minimal Geodata Models (MGM) cite:[mgm] using the Swiss INTERLIS modeling language. It is also mandatory for these models to provide styling and symbology instructions according to a spreadsheet-based model (e.g. https://www.bafu.admin.ch/bafu/en/home/state/data/geodata-models/water--geodata-models.html).



Modell	Topic	Klasse	Geometrie-Attribut	Geometrie-Typ	Attribut-Abhängigkeit	Stil-ID	Experten-Stil	Bemerkungen	Massstabbhänigkeit		Darstellungs-reihenfolge
									Massstab (optional; Bodenmass; Default Meter)		
Modellname (mit Komma getrennt, falls mehrere Modelle involviert)	Topicname (optional)	Klassen-, Tabellen- oder Viewname ("Simple Feature Class")	Geometry-Attribut-Name (existieren mehrere Geometrie-Attribute in der Klasse, ist der Name zwingend anzugeben)	Punkt (P), Linie (L), Polygon (A), Raster (R)	Filterkriterien (optional; logischer Ausdruck, Teilmenge von CQL)	ID (Referenz) einer Stil-Definition (eindeutig innerhalb Darstellungsmodell)	UserStyle Name in separatem SE 1.1 Dokument (optional; wenn benutzt, ist "Stil-ID" leer)	Thumbnail-Illustration oder andere informelle Bemerkungen (optional)	Minimum	Maximum	Von hinten (1) nach vorne (optional; Default=1)
[Text]	[Text]	[Text]	[Text]	[P, L, A, R]	[Text]	[Text]	[Text]	[Text]	[Zahl]	[Zahl]	[Zahl]
Gewaesserraum_LV95_V1	GewR	GewR	Geometrie	A	Verzicht = FALSE	A-KeinVerzicht					1
Gewaesserraum_LV95_V1	GewR	GewR	Geometrie	A	Verzicht = TRUE	A-Verzicht					
Gewaesserraum_LV03_V1	GewR	GewR	Geometrie	A	Verzicht = FALSE	A-KeinVerzicht					
Gewaesserraum_LV03_V1	GewR	GewR	Geometrie	A	Verzicht = TRUE	A-Verzicht					

Figure 3. Styling and symbology instructions according to a spreadsheet-based model (from the Area reserved for water MGM)

Example 1. Geocss styling instructions for Area reserved for water MGM

```
/* Metadata
* @title GeoCSS encoding for GeoServer
* @abstract Representation model
*/
/* Main rule*/
*{
  /* @title ERE */
  [obligation = 1 ] {
    fill:#ddebf7;
      stroke:#ffcc00;
```

```
      stroke-width: 6px;
    };
  /* @title Renonciation */
  [obligation = 0 ] {
      stroke:#ffcc00;
      stroke-width: 4px;
        stroke-dasharray: 4 4;
    };
}
```

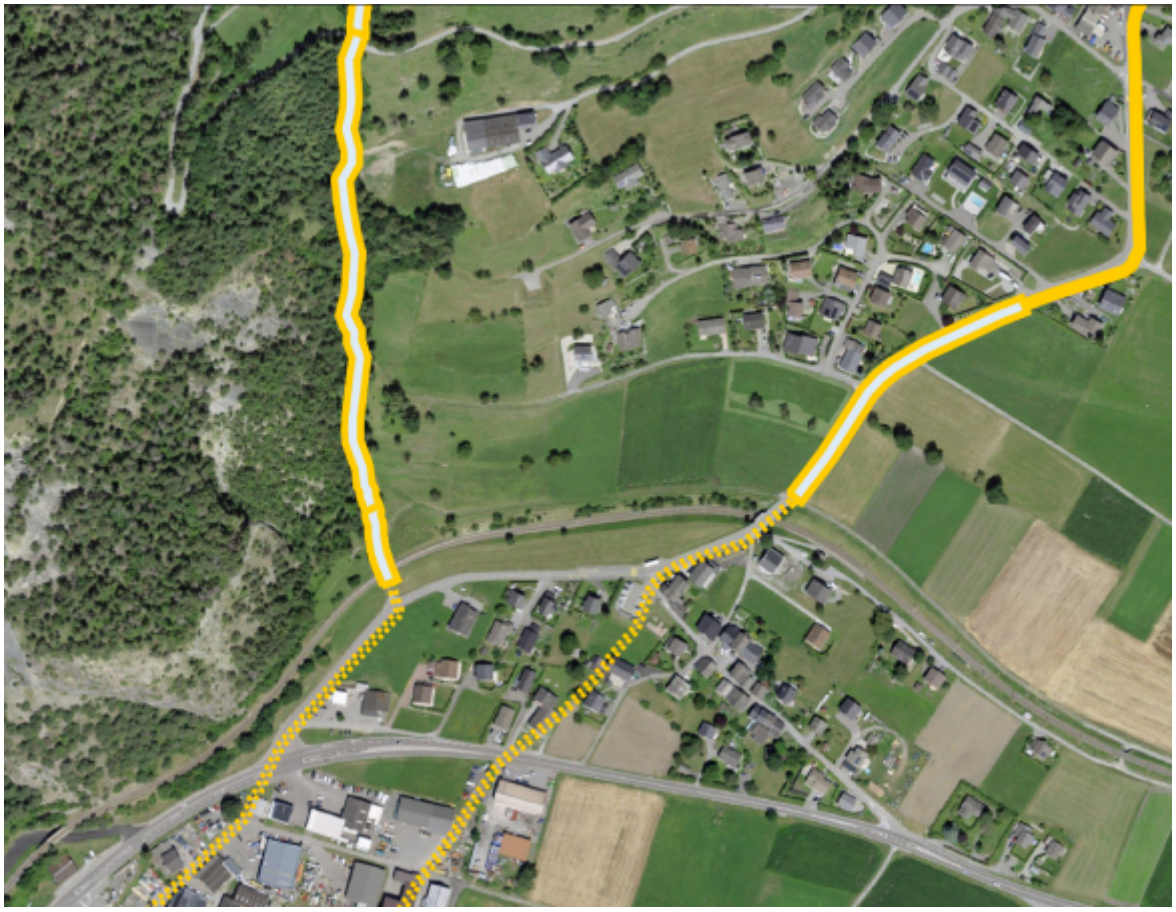


Figure 4. Overview of the Area reserved for water portrayal using the above GeoCSS encoding for GeoServer

First elements to highlight

Regarding the publication of vector data using the OGC API Features standard, we can state that all software packages already support this standard cite:[ogc-api-feature-implementations]. Regarding the tiling of data sets, for a long time the existing WMTS standard has been largely used, but a standard for vector tiling has never been established up to now. A possible explanation for this lack of standardization is on the one hand the complexity of vector tiling (e.g. regarding the handling of attributes or projections), but on the other hand the success of the Mapbox Vector tiles specification cite:[mvt-spec] that have been widely adopted. The OGC API Tiles specification is on a conceptual level similar to the WMTS standard and defines the addressing and tiling of the data. One difference is that the OGC API Tiles specification allows for several formats (both vector and raster) to be computed. This way of defining tiles assures on the one hand the compatibility with existing WMTS services (i.e. allowing applications to easily integrate both existing WMTS layers with tiled

vector layers), but also with the Mapbox Vector tiles specification. On the software side GeoServer already supports the OGC-API tiles specification rendering the formats jpg, png, GeoJSON, topojson and mapbox-vector-tile.

Concerning portrayal, we may notice two related aspects: about OGC API Styles, about OGC SymCore. Firstly, OGC API Styles is inline with the conceptual model for styles, style encodings and style metadata as documented in chapter 6 of the OGC Testbed-15: Encoding and Metadata Conceptual Model for Styles Engineering Report. Especially it states that a style may be made available in one or more so-called stylesheets. Moreover style metadata are made available through the API with general descriptive information about the style, structural information (e.g., layers and attributes), and so forth to allow users to discover and select existing styles for their data. Having several stylesheets available does not guarantee the same visualization of the cartographic result for the final user, because each stylesheet may be based on different models and encodings (e.g. SLD, Mapbox style, GeoCSS, etc). Nonetheless, it opens the possibility to make full use of the cartographic capabilities and richness of the various underlying symbology models.

Secondly, OGC SymCore pushes forward portrayal interoperability with the idea to standardize also the symbology part. The approach is so-called one conceptual model, many encodings, which means that many flavors of encodings are possible but each in conformance with a common conceptual rendering behavior of cartographic capabilities. The intent is that finally, independently of the compliant encoding used, the cartographic result will be the same for the final user.

Work Package 4 (WP4)

The Work Package 4 (named hereafter WP4) has the intent to evaluate…

Conclusion

Lorem ipsum.

Conferences

- Spiergarten Treffen 2022
- FOSS4G 2022

Follow up meetings

25-05-2022

Date & Time

- 25.05.2022 09:00-10:00

Attendees

- Frank
- Jens
- Maxime

Agenda

Who	What	Comments
Frank	Plan MS-Teams meetings every 3 weeks	Done
Frank	Synchronisation of meeting notes on GitHub	Done
Frank	Valuation of presentation materials, publication etc. on GitHub	Done
Frank	Add a link to the TB platform on the project documentation an ad hoc section in the documentation	Done

Related documents and resources

- Project Documentation
- NGDI-20-60 OGC API Testbed Platform

Next meeting

31.05.2022

Miscellaneous

Nothing to mention

31-05-2022

Date & Time

31.05.2022 09:00-12:00

Attendees

- Frank
- Maxime

Agenda

Who	What	Comments
Frank	Prepare templates for meeting notes on GitHub	Done
Frank	Feed the documentation and the testbed platform (Howto's, WP's Results)	To be carried out throughout the project
Frank	Depict the CH Standards Landscape - presentation of the situation as it is	Done
Frank	Make a decision matrix for the choice of standards (OGC, eCH). Typical questions & critical factors are: • When is a standard interesting / ready to be standardized in Switzerland as well? • Groundtruthing/verification (Interviews with e-CH / users etc.)	To be carried out throughout the project

Related documents and resources

- Nothing to mention

Next meeting

20.06.2022

Miscellaneous

Nothing to mention

20-06-2022

Date & Time

20.06.2022 09:00-10:00

Attendees

- Frank

- Jens
- Maxime
- Olivier

Agenda

Who	What	Comments
Olivier	Presentation of the FOSS4G publication	Done
Jens	Discusses that the new OGC standards are shifting focus towards web-applications. This trend potentially creates problems regarding bridges between INTERLIS and new OGC standards.	-
Jens	Proposes the build the output in conformity with eCH-structure (e.g. https://www.ech.ch/it/standards/60396)	This idea is approved and has to be added to the project final restitution.
Jens	Proposes to implement APIs in relation to MeteoSwiss data (following Massimiliano's idea)	This is a great idea for a future iteration.
Frank	State of play of financial resources	Complete the financial report for the 24.06.2022
Frank	Dissemination of the work in the context of workshops	Frank will take care of the exchange with swisstopo regarding the organisation of the Kolloquium

Related documents and resources

- FOSS4G article
- 20220620-NGDI-20-60 follow up meeting notes

Next meeting

11.07.2022

Miscellaneous

Nothing to mention

11-07-2022

Date & Time

11.07.2022 09:00-09:30

Attendees

- Frank
- Maxi
- Maxime

Agenda

Who	What	Comments
Frank	eCH-0056 kickoff meeting in august	
Maxime	integrate the ech-0056 structure into the report	
Frank	will provide a proposition of reporting structure by the end of the week (15/07/2022)	
Maxi	SOS mentioned in ecH-0056 but no recommendation made	
Maxi	MeteoSwiss not really interested in new OGC APIs but other data provider could be identified	
Frank	Quick How to’s to play with the new OGC APIs (e.g. ArcGIS, QGIS, web client, etc.)	

Related documents and resources

Next meeting

01.08.2022

Miscellaneous

Nothing to mention

01-08-2022

Date & Time

01.08.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

22.08.2022

Miscellaneous

22-08-2022

Date & Time

22.08.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

12.09.2022

Miscellaneous

12-09-2022

Date & Time

12.09.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

03.10.2022

Miscellaneous

03-10-2022

Date & Time

03.10.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

24.10.2022

Miscellaneous

24-10-2022

Date & Time

24.10.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

14.11.2022

Miscellaneous

14-11-2022

Date & Time

14.11.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

05.12.2022

Miscellaneous

05-12-2022

Date & Time

05.12.2022 09:00-10:00

Attendees

Agenda

Who	What	Comments

Related documents and resources

Next meeting

tbd?

Miscellaneous

References

bibliography::[]